

Supplementary Material: Leverage-Driven Software Architecture: A Probabilistic Framework for Architectural Decision-Making

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1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

ID	Lean Handle / Source	ID	Lean Handle / Source
L9	ACS1 paper1/HandleAliases.lean	L10	ACS2 paper1/HandleAliases.lean
L11	ACS3 paper1/HandleAliases.lean	L12	ACS4 paper1/HandleAliases.lean
L13	ACS5 paper1/HandleAliases.lean	L14	ACS6 paper1/HandleAliases.lean
L15	ACS7 paper1/HandleAliases.lean	L16	ACS8 paper1/HandleAliases.lean
L17	ACS9 paper1/HandleAliases.lean	L18	ADV1 paper1/HandleAliases.lean
L19	ADV2 paper2/Ssot/HandleAliases.lean	L20	ADV3 paper2/Ssot/HandleAliases.lean
L21	ADV4 paper2/Ssot/HandleAliases.lean	L22	ADV5 paper2/Ssot/HandleAliases.lean
L23	AFM1 paper2/Ssot/HandleAliases.lean	L24	AFM10 paper2/Ssot/HandleAliases.lean
L25	AFM11 paper2/Ssot/HandleAliases.lean	L26	AFM12 paper2/Ssot/HandleAliases.lean
L27	AFM13 paper2/Ssot/HandleAliases.lean	L28	AFM14 paper2/Ssot/HandleAliases.lean
L29	AFM15 paper2/Ssot/HandleAliases.lean	L30	AFM16 paper2/Ssot/HandleAliases.lean
L31	AFM17 paper2/Ssot/HandleAliases.lean	L32	AFM18 paper2/Ssot/HandleAliases.lean
L33	AFM19 paper2/Ssot/HandleAliases.lean	L34	AFM2 paper2/Ssot/HandleAliases.lean
L37	AFM5 paper2/Ssot/HandleAliases.lean	L38	AFM6 paper2/Ssot/HandleAliases.lean
L39	AFM7 paper2/Ssot/HandleAliases.lean	L40	AFM8 paper2/Ssot/HandleAliases.lean
L41	AFM9 paper2/Ssot/HandleAliases.lean	L42	ALT1 paper1/HandleAliases.lean

2 Claim-to-Lean Handle Mapping

This section maps each paper claim to its corresponding Lean formalization.

Paper claim	Lean handle
Theorem 4.4: Leverage Composition	L18, L19
Theorem 4.2: Leverage-Error Tradeoff	L29
Theorem 4.1: Leverage Maximization Principle	L28, L31
Theorem 4.3: Optimal Architecture	L34, L38

Auto summary: mapped 4/4 (full=4, derived=0, unmapped=0).

3 Proof Hardness Index

Paper handle	Hardness profile	Regime tags	Lean support
thm:composition	unspecified	-	L18, L19
thm:leverage-error	unspecified	-	L29
thm:leverage-max	unspecified	-	L28, L31
thm:optimal	unspecified	-	L34, L38

Auto summary: indexed 4 claims by hardness profile (unspecified=4).

4 Assumption Ledger

4.1 Assumption Ledger (Auto)

Source files.

- Leverage/Examples.lean
- Leverage/Foundations.lean
- Leverage/Probability.lean
- Leverage/Theorems.lean
- Leverage/WeightedLeverage.lean

Assumption bundles and fields.

- (No assumption bundles parsed.)

Conditional closure handles.

- (No conditional theorem handles parsed.)

First-principles Bayes derivation handles.

- (No Bayes-derivation theorem handles found.)

Compact Bayes handle IDs.

- (No compact Bayes alias IDs found.)